

Unit

Graph

What is graph?

Graph:

Edge: edge is a connection b/w 2 vertices.

⇒ It is a pair $G=(V, E)$

Where

$V = V(G) =$ set of vertices

$E = E(G) =$ set of edges

Vertices: Vertices are simple objects.

Example:

$V = \{s, u, v, w, x, y, z\}$

$E = \{(x, s), (x, v), (x, v)_2,$

$x, u), (v, w), (s, v), (s, u),$

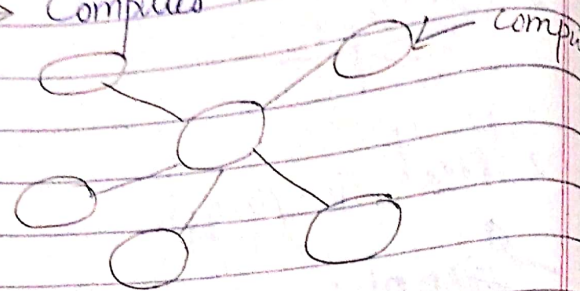
$(s, w), (s, y), (w, y), (u, y), (u, z),$

$(y, z)\}$

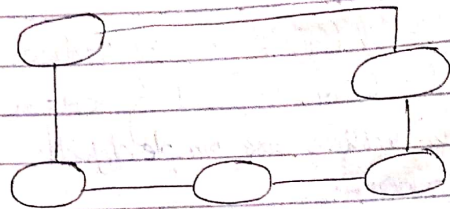
Application of

Graph

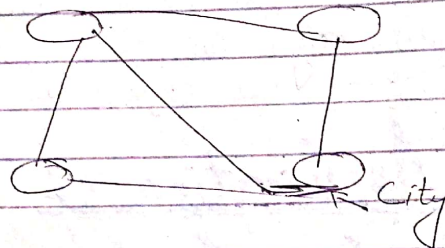
⇒ Computers networks



⇒ Electrical circuits
Resistor/inductor....



⇒ Road map

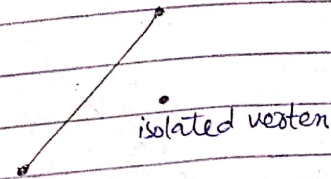


Edges:

⇒ An edge may be labeled by a pair of vertices, for instance $e = (v, w)$.

⇒ e is said to be incident on v and w .

⇒ Isolated vertex = A vertex without incident edges.

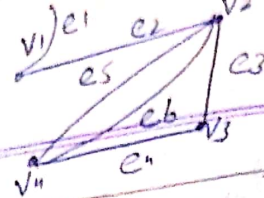


Special edges:

Parallel edges:

⇒ Two or more edges joining a pair of vertices

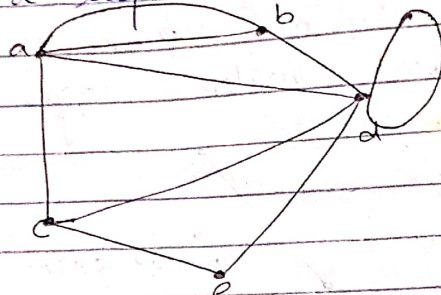
⇒ In the example a and b are joined by two parallel edges.



Loops:

⇒ An edge that starts and ends at the same vertex is called loop.

⇒ In example vertex d has a loop.



⇒ Also called self edges (u, u) .

Graph Categorization

Undirected graph:

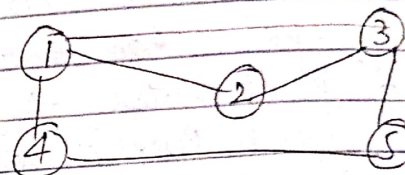
⇒ An undirected graph is a graph where the edges have no directions.

⇒ The edges in an undirected graph are called undirected edges.

⇒ For example, $G=(V, E)$ where

$V = \{1, 2, 3, 4, 5\}$ and

$E = \{(1, 2), (1, 3), (1, 4), (2, 3), (3, 5), (4, 5)\}$



Degree of a vertex

⇒ In an undirected graph, the no of edges incident to the vertex.

⇒ Degree of vertex v is denoted by $\deg(v)$.

⇒ For example,

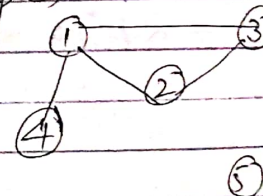
$$\deg(1) = 3$$

$$\deg(2) = 2$$

$$\deg(3) = 2$$

$$\deg(4) = 1$$

$$\deg(5) = 0$$



What are isolated & pendant vertices?

Isolated vertex

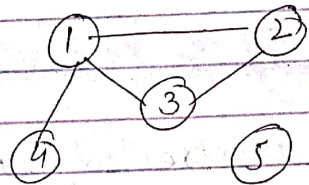
→ A vertex of degree zero is called isolated vertex.

→ For example, vertex 5 is isolated.

Pendant vertex

→ A vertex of degree 1 is called pendant vertex.

For example, vertex 4 is pendant vertex.



Hand Shaking Theorem:

→ Let $G=(V,E)$ be an undirected graph. Then
 $2|E| = \sum \deg(v), \forall v \in V$

Example: How many edges are there in a graph with 10 vertices each of degree six?

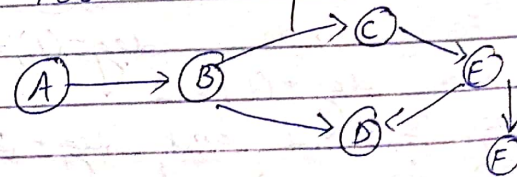
Because the sum of degree of vertices is $6 \times 10 = 60$, it follows that $2|E| = 60$, Therefore, $|E| = 30$.

Directed Graph:

→ A directed graph or digraph is a graph where each edge has a direction.

→ The edge in a digraph are called arcs or directed edges.

→ For example,



→ degree of vertex in digraph:

→ In diagram, there are two types of degree of a vertex

→ in-degree

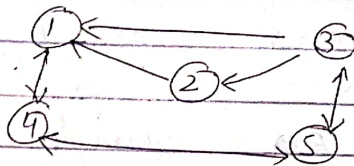
→ out-degree

→ In degree: In degree of a vertex v , denoted by $\deg^-(v)$,

→ Out degree: out degree of a vertex v , denoted by $\deg^+(v)$.

→ In degree: Is the no of edges entering the vertex in a digraph.

→ out degree: Is the no of edges leaving the vertex.



→ For example, $\deg^-(1) = 3$ and $\deg^+(1) = 1$

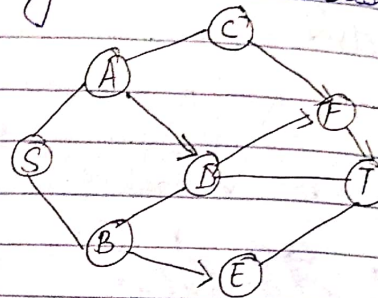
$\deg^-(2) = 1$ and $\deg^+(2) = 1$

→ for a digraph:

$$|E| = \sum \deg^-(v) + \sum \deg^+(v), \forall v$$

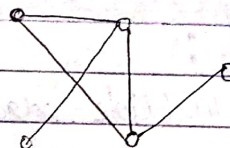
Mixed graph:

→ A mixed graph is a graph in which some edges are directed and some may be undirected.



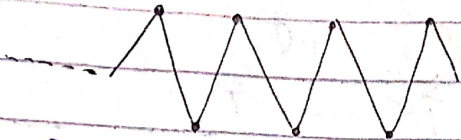
Finite Graph:

→ A finite graph is a graph in which the vertex set and the edge set are finite sets.



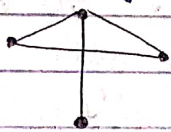
Infinite Graph:

→ A infinite is a graph in which the vertex set and the edge set are infinite sets.



Simple Graph:

→ A simple graph is an undirected graph without multiple edges or loops.

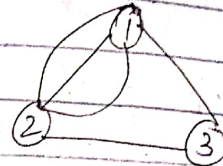


Simple graph

Multigraph:

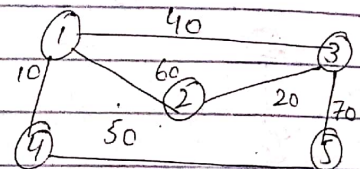
→ multiple edges are two or more edges that connect the same two vertices.

→ If the same pair of vertices have more than one edges (multiple edges), that graph is called a multigraph.



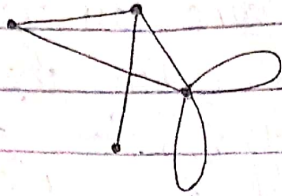
Weighted Graph:

→ A weighted graph is a graph where all edges are assigned weights.



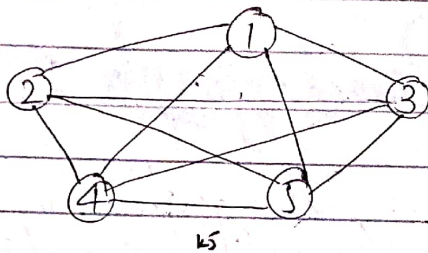
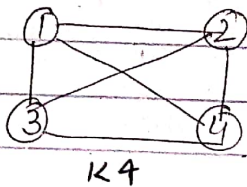
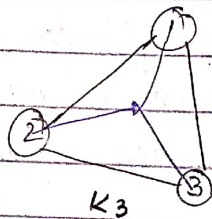
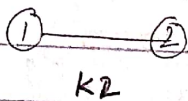
Pseudograph:

→ A graph in which loops and multiple edges are allowed is called pseudograph.



Complete graph:

→ A simple graph G is said to be complete if every vertex in G is connected with every other vertex.



$$= \frac{n(n-1)}{2}$$

n = no of vertices

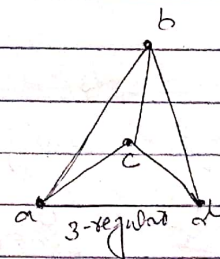
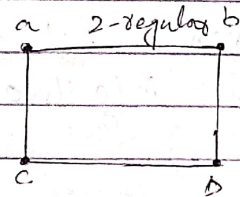
Regular graph:

→ A graph in which all the vertices are of equal degree is called a regular graph.

Cubic graph:

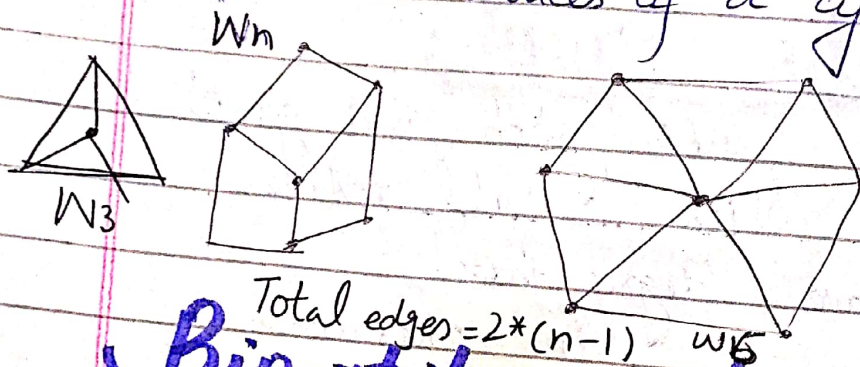
→ Cubic graph is a graph in which all vertices have degree three.

In other words a cubic graph is a 3-regular graph. e.g. Petersen graph C_n .



Wheel Vertex graph:

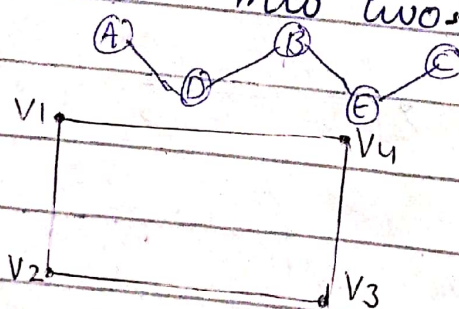
→ A wheel graph is a graph formed by connecting a single universal vertex to all vertices of a cycle.



Total edges = $2 \times (n-1)$

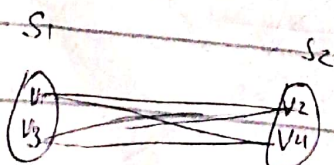
Bipartite graph:

→ A bipartite graph is a graph in which all the vertices can be partitioned into two non-empty disjoint sets such that



$V = V_1 \cup V_2$ and $V_1 \cap V_2 = \emptyset$

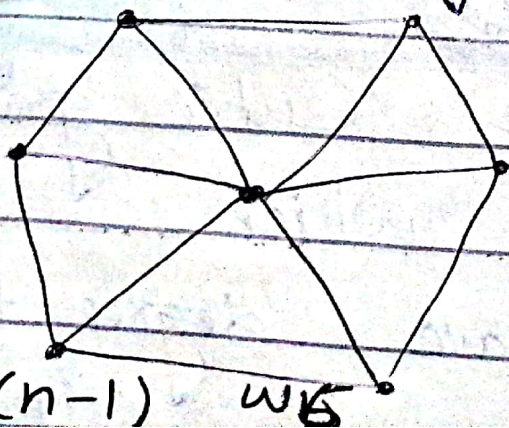
$S_1 = \{V_1, V_3\}$, $S_2 = \{V_2, V_4\}$



e.g. C₆ is a bipartite graph.

Theorem:

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.



$$= 2 * (n-1) \quad \text{w.b.}$$

Matching:

- pair two vertices from 2 sets.
- applications to n jobs

$\Rightarrow$ Matching in a graph is a subset of edges that no two edges share a vertex.

$\Rightarrow$ Every vertex degree should be $\deg(v) \leq 1$

$\Rightarrow$ Perfect matching $\deg(v) = 1$ of every vertex.

Maximum matching:-
- matching with the largest no of edges.

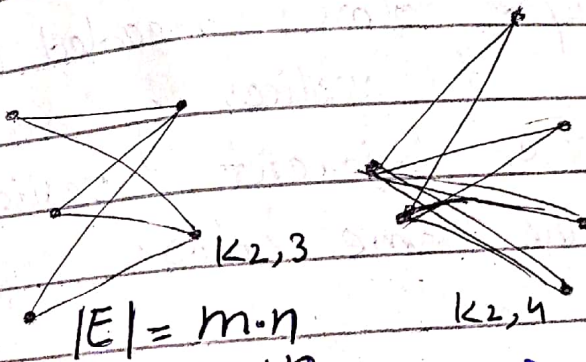
vertices in set one should connect the vertices of 2nd set but cannot connect with each other of the set one.

Such that every edge in the graph connects in V_1 and a vertex in V_2 .

Complete bipartite graph

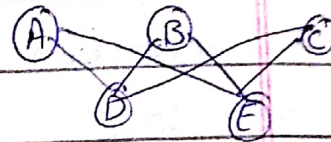
→ Also called bicoloured graph

→ A complete bipartite graph is a bipartite graph in which each vertex in the first set is joined to every single vertex in the second set. $K_{m,n}$



$$|E| = m \cdot n$$

$$|V| = m + n$$



$$V_1 = \{A, B, C\}$$

$$V_2 = \{D, E\}$$

$$V_1 \cup V_2 = V$$

$$V_1 \cap V_2 = \phi$$

$$V = \{A, B, C, D, E\}$$

Cyclic Graph:

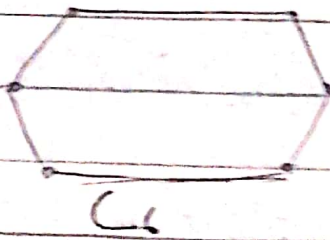
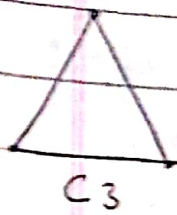
→ A cyclic graph is a graph containing at least one graph cycle.

→ A graph that is not cyclic is said to be an acyclic graph.

→ A cyclic graph possessing exactly one cycle is called a unicyclic.

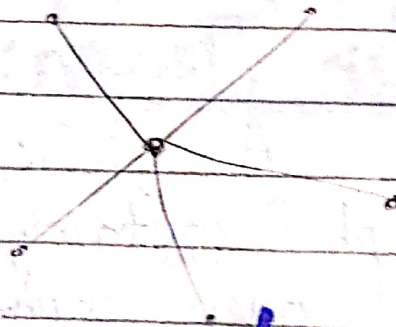
$$C_n, n \geq 3$$

vertices and edges $\{1,2\}, \{2,3\}, \dots, \{n-1,n\}$
and $\{n,1\}$



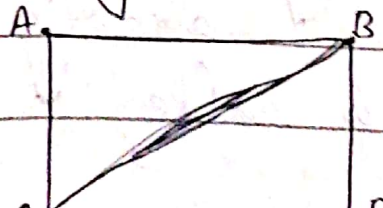
Chromatic no of graph:

→ The chromatic number of a graph is the smallest number of colors needed to color the vertices of so that no two adjacent vertices share the same color.



Planar graph:

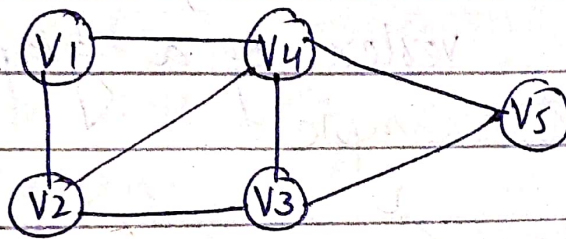
→ A graph which has no cross edges.



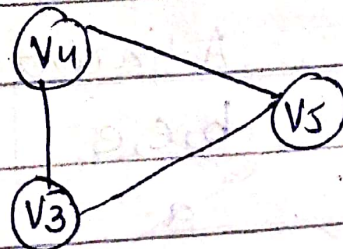
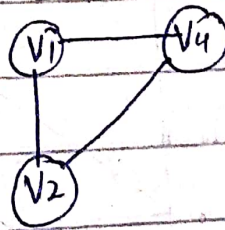
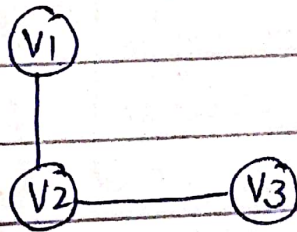
Subgraph:

→ A sub-graph of graph G is a graph G' such that $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$.

In the following graph:



Some possible subgraphs are:

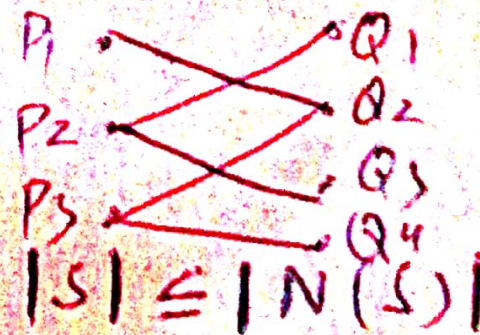


Hall's Marriage

Proved in
1935 by Philip Hall

Theorem

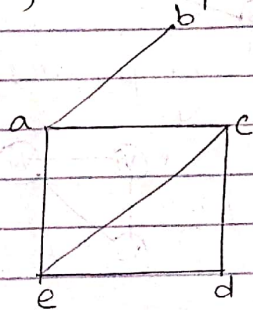
The bipartite graph $G=(V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1 .



Graph Representation

i) Adjacency Lists:

→ It is a way to represent a graph without multiple edges. And it specifies all the vertices that are adjacent to each vertex of a graph.
for example



Vertex	Adjacent vertices
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

⇒ is a 2D array of size $V \times V$

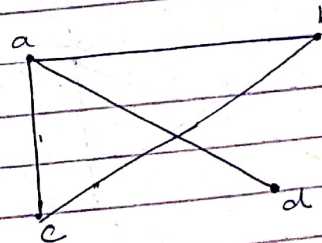
ii) Adjacency matrices:

→ $A(A_G)$ is a $n \times n$ zero-one matrix with 1 as its $(i, j)^{\text{th}}$ entry when i and j are adjacent and 0 otherwise.

$$A[a_{ij}] = \begin{cases} 1 & \text{if } (i, j) \text{ is edge of } G \\ 0 & \text{otherwise} \end{cases}$$

Represent following graph with use of adjacency matrix.

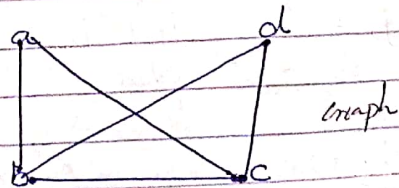
	a	b	c	d
a	0	1	1	1
b	1	0	0	0
c	1	0	0	0
d	1	0	0	0



Draw a graph with Adjacency matrix.

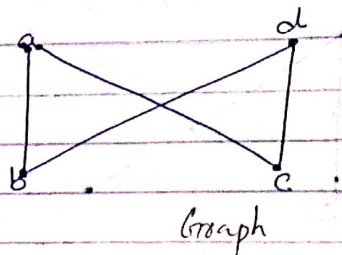
a)

	a	b	c	d
a	0	1	1	0
b	1	0	1	1
c	1	1	0	1
d	0	1	1	0



b)

	a	b	c	d
a	0	1	1	0
b	1	1	0	0
c	1	1	0	0
d	0	1	1	0



Incidence matrices

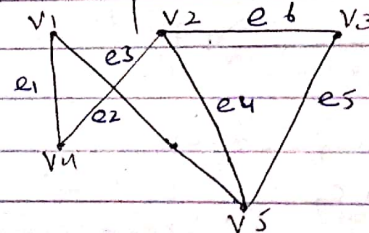
→ Let $G=(V,E)$ is an undirected graph.

Suppose that $v_1, v_2, v_3, \dots, v_n$ are the vertices and $e_1, e_2, e_3, \dots, e_n$ are the edges of G .

The incidence matrix with respect to this ordering of V and E is the $n \times n$ matrix $M=[m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$

example:



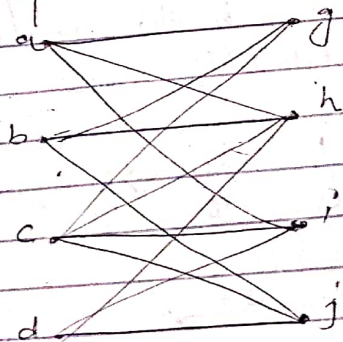
Sol

	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	0	1	1	0	1
v_3	0	0	0	0	1	1
v_4	1	0	1	0	0	0
v_5	0	1	0	1	1	0

Isomorphic Graphs:

- Two graphs and are said to be isomorphic if
- $\rightarrow f$ is one-to-one and onto
 - $\rightarrow$ Their number of components (vertices and edges) are same
 - $\rightarrow$ Their edge connectivity is retained.

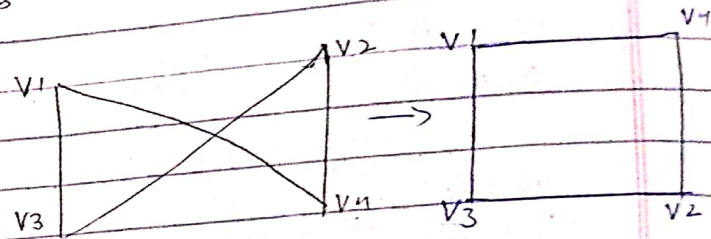
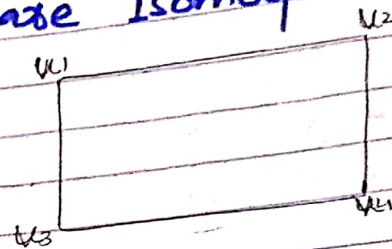
Example



Properties:

- $\rightarrow$ Same no of edges
- $\rightarrow$ Same no of vertices
- $\rightarrow$ Equal no of vertices with given degree
- $\rightarrow \cong$ represents isomorphism graph.

Show that G and H are Isomorphic.



$$\left. \begin{array}{l} f(u_1) = v_1 \\ f(u_2) = v_4 \\ f(u_3) = v_3 \\ f(u_4) = v_2 \end{array} \right\} \text{one to one function mapping}$$

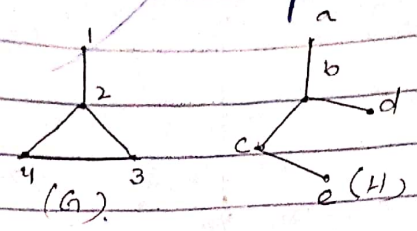
$$\left. \begin{array}{l} (u_1, u_2) \cong (v_1, v_4) \\ (u_1, u_3) \cong (v_1, v_3) \\ (u_3, u_4) \cong (v_3, v_2) \\ (u_4, u_2) \cong (v_2, v_4) \end{array} \right\} \begin{array}{l} G \text{ and } H \text{ are} \\ \text{Isomorphic} \end{array}$$

Application of graph Isomorphism:

- $\rightarrow$ Chemistry
- $\rightarrow$ bioinformatics
- $\rightarrow$ Design of electronic circuits
- $\rightarrow$ computer vision

Check whether G and H are isomorphic.

a)



$NG = 4$
 $NH = 4$

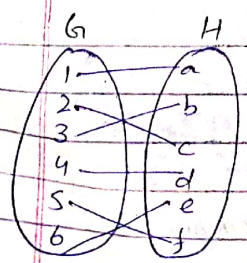
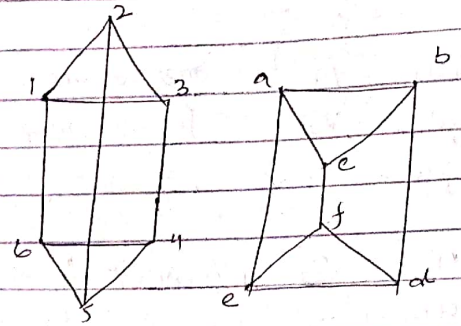
} no of edges

$\Rightarrow G$ and H are not isomorphic

$VG = 4$
 $VH = 5$

} no of vertices

(b)



G_1 H
 $(1,2) \cong (a,c)$
 $(1,3) \cong (a,b)$
 Definition:-
 $\Rightarrow G_1$ and G_2 are isomorphic
 if their vertices can be ordered
 in such a way that the
 adjacency matrices M_{G_1} and M_{G_2}
 are identical.
 $\Rightarrow$ function of isomorphic graph like $f(G)$ is

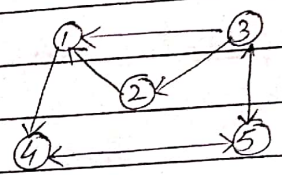
Date _____

Define Path.

Path:

$\Rightarrow$ A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along the edges of the graph.

$\Rightarrow$ There can be more than one path between two vertices.



$\Rightarrow$ Paths from 5 to 1 are 5,3,1 and 5,3,2,1

$\Rightarrow$ A simple path is a path where all the vertices are distinct.

$\Rightarrow$ For example, 1,4,5,3 is a simple path, but 1,4,5,4 is not.

$\Rightarrow$ Length of a path is the no of edges considered in the path.

$\Rightarrow$ Length of the path 1,4,5,3 is three.

Date _____

Day **MTW**

Define circuit.

Circuit:

- A circuit is a path whose first and last vertices are the same.
- For example, the path 3, 2, 1, 4, 5, 3 is a circuit.

Define cycle.

Cycle:

- A path of length > 0
- which begins & ends at the same vertex
- all edges & non-ending vertices are distinct
- For example, 1, 4, 5, 3, 1 is a cycle, but 1, 4, 5, 4, 1 is not a cycle.

In a graph the sum of the degrees of the vertices equals twice the number of edges. Why?

Because each one edge is incident on 2 vertices.

Day **MTWTF**

Euler Paths & Circuits

Euler path: (vertex can be repeated but edge can't)

- A path including every edge exactly once & different first and last vertices.

Euler circuit:

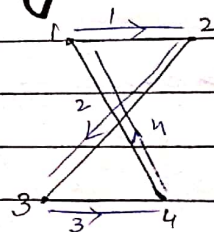
- A path including every edge exactly once & the same first and last vertices.

Euler Graph:

- A Graph that contains an Euler path is called as Euler graph.

- Euler Graph is always connected, because Euler path contains all the edges of the graph.

e.g



1 → 2 → 3 → 4 is Euler path.

1 → 2 → 3 → 4 → 1 is Euler circuit.

In this each edge is only visited once.

Date _____

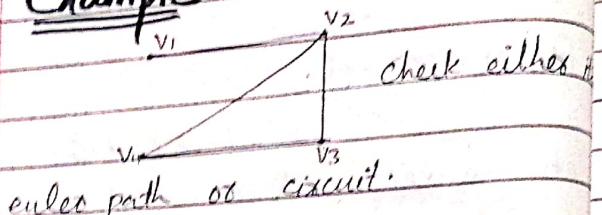
Note ≡

→ A connected graph is Euler graph iff it has at most 2 odd degree vertices.

Euler circuit → Each vertex is of even degree

Euler path → max two vertices odd degree.

Example:



Solution:

→ This is Euler path because we visited all edges only once.

→ This is not Euler circuit because initial and last vertex are not same.

Degree of vertices:

→ $d(v_1) = 2 \rightarrow d(v_2) = 2 \rightarrow d(v_3) = 3 \rightarrow d(v_4) = 1$
one vertex degree is odd.

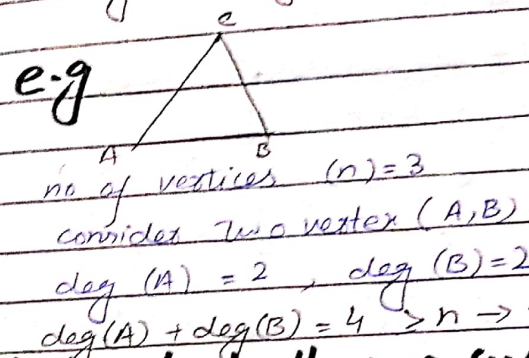
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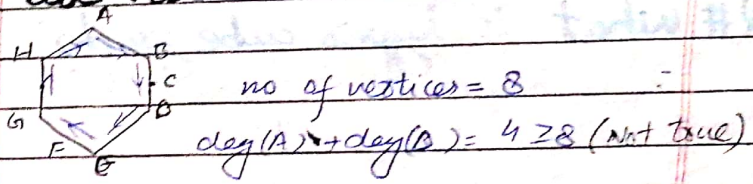
→ Length of H. path in a connected graph of n vertices is $(n-1)$ edges.

Theorem: Ozer's theorem

Let G be a graph of n vertices then G has a H-path if for only two vertices u and v of G
 $\deg(u) + \deg(v) \geq n$

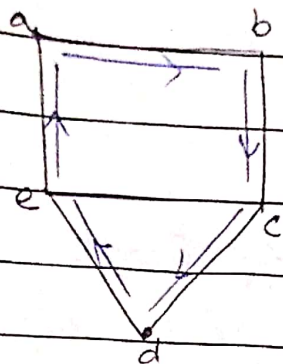


Let's check theorem sufficient are not:



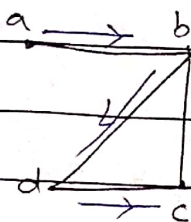
Hence, this condition does not true.

a)



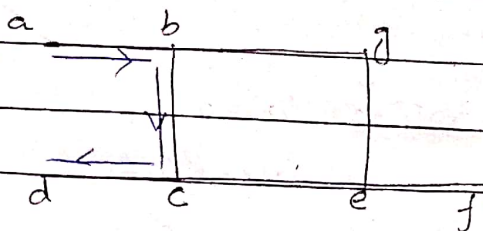
H. circuit (a, b, c, d, e, a)

b)



H. path [a, b, d, c]

c)



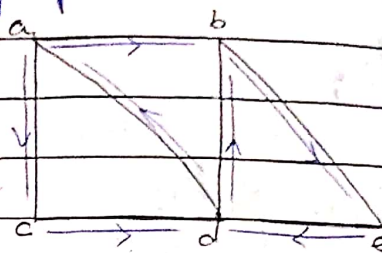
not contain either path or circuit

Q# what is hypercube graph

→ The hypercube or n-cube graph is a graph with 2^n vertices each represented by a n-bit string.

→ A hypercube graph of 2^n vertices is denoted by Q_n and total number of edges $n \times 2^{n-1}$.

Find Euler path for graph below.



a → c → d → b → e → d → a → b

Degree of vertices:

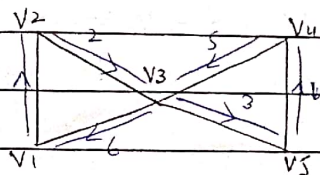
→ $d(a) = 3$ ∴ 2 vertex

→ $d(b) = 3$ with odd degree.

→ $d(c) = 2$

→ $d(d) = 4$

→ $d(e) = 2$



initial vertex = V_1

final vertex = V_1

This is euler circuit $V_1 \rightarrow V_2 \rightarrow V_3$

→ $V_5 \rightarrow V_4 \rightarrow V_1$

Degree of vertices:

$V_1 = 2$, $V_4 = 2$

$V_2 = 2$, $V_5 = 2$

$V_3 = 4$

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Hamiltonian Path & Circuit

Circuit:

William Rowan Hamilton

- In 1857 the Irish mathematician

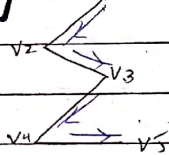
Hamiltonian path:

→ A path including every vertex exactly once

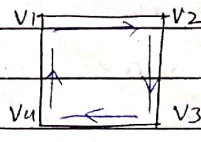
Hamiltonian circuit:

→ A path including every vertex exactly once and first and last vertex are same.

all the edges may or may not be covered
e.g. V_1 but edges must not repeat

H. path $\rightarrow V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_4 \rightarrow V_5$

e.g.

H. circuit $\rightarrow V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_4 \rightarrow V_1$

Note =

→ on removing only one edge from a Hamiltonian circuit, then we are left with Hamiltonian Path.

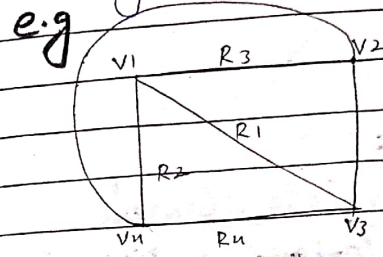
Kuratowski's Theorem:

Every non-planar graph has a subgraph that is homeomorphic to K_5 or $K_{3,3}$.

Kuratowski's graph:

Two specific non-planar graphs, are called Kuratowski's graph.

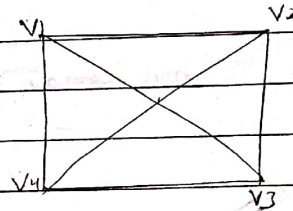
e.g. K_5 is Kuratowski's graph drawn in the plane without any edges crossing.



Non-planar: Kur

→ Graph that cannot be drawn on a plane without a crossing b/w its edges is called Non-planar graph.

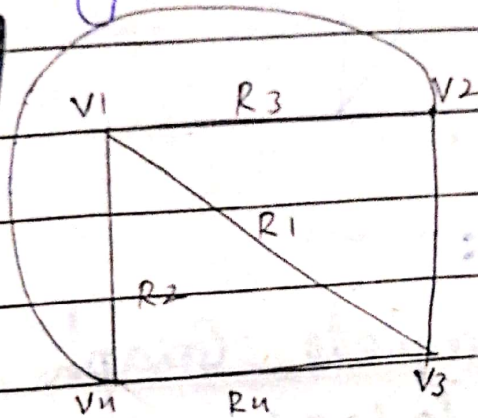
e.g.



Planar Graph

→ A Graph is called planar if it can be drawn in the plane without any edges crossing.

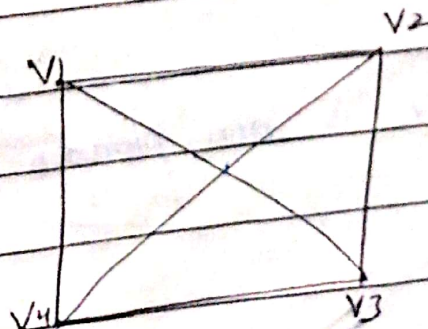
e.g



Non-planar: K₄

→ Graph that cannot be drawn on a plane without a crossover b/w its edges is called Non-planar graph.

e.g



Kuratowski's Theorem:

Every non-planar graph has a subgraph that is homeomorphic to K_5 or $K_{3,3}$

Kuratowski's graph:

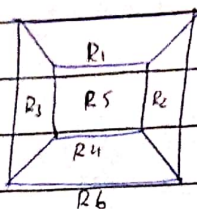
Two specific non planar graphs, are called Kuratowski's graph:

e.g. K_5 is Kuratowski's graph.

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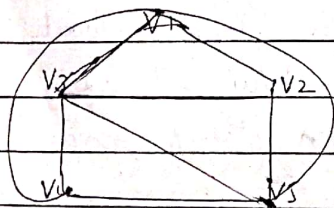
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Convert this graph into planar.



Theorem:

A complete graph of five vertices is non-planar.

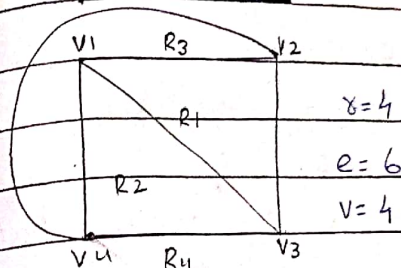


Hence it is non-planar graph.

Euler's formula:

Let G be a connected planar simple graph with e edges and vertices v . Let r be the no. of regions in planar rep G then,

$$r = e - v + 2$$



$$r = 4$$

$$e = 6$$

$$v = 4$$

$$4 = 6 - 4 + 2$$

$$4 = 2 + 2$$

$$4 = 4$$

Q# : if there are 20 vertices each of degree 3, then into how many regions does a rep of this planar graph splits the plane.

$$V = 20, \sum \deg(V) = 20 \times 3 = 60$$

$$2e = 60 \Rightarrow e = 30 \therefore \sum \deg(V) = 2e$$

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$$\delta = e - v + 2$$

$$= 30 - 20 + 2$$

$$= 10 + 2$$

$$\delta = 12$$

Prove Euler's formula using Mathematical induction.

i) Basic step: $P(1) \rightarrow \text{True}$

$$n=1 \quad \text{---} \quad \delta=1$$

$$\delta_1 = e_1 - v_1 + 2$$

$$= 1 - 2 + 2$$

$$\delta_1 = 1 = \delta \quad \text{True}$$

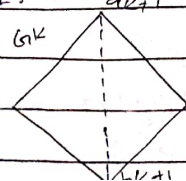
ii) Induction step (assume that above result is true for k)
for $n=k$

$$\delta_k = e_k - v_k + 2 \text{ is true for } G_k$$

iii) Verification for $n=k+1$

Let (a_{k+1}, b_{k+1}) be the edge that is added to G_k .

Case 1:



$$\delta_{k+1} = \delta_k + 1$$

$$e_{k+1} = e_k + 1$$

$$v_{k+1} = v_k \because \text{Both vertices are in } G_k$$

Note =

$\rightarrow$ A connected graph G_1 may have at most $(n-2)$ cut vertex. (n is no. of vertices)

$\rightarrow$ The maximum no. of cut edges possible is $(n-1)$.

$\rightarrow$ whenever cut-vertex exist then cut edge may or may not exist.

$\rightarrow$ if cut edge exist cut vertex also exist because at least one vertex of a cut edge is a cut vertex.

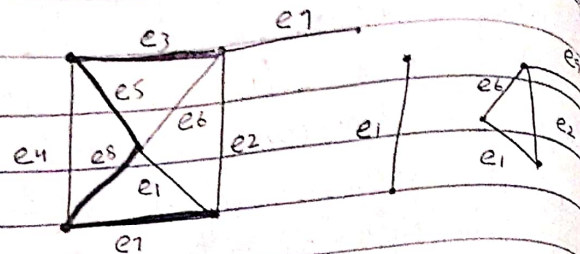
Cut set of a graph:

$\rightarrow$ Let $G = (V, E)$ be a connected graph.

A subset E' of E is called cut set of G . If deletion of all the edges of E' from G makes G disconnected.

Example:

Here cut set is $E_1 = \{e_1, e_2, e_3, e_4\}$



Diff b/w

Adjacent

→ a vertex connected to a vertex.

Incident

→ a vertex connected to an edge.

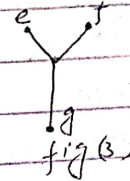
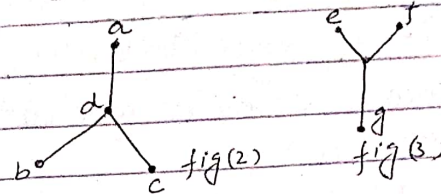
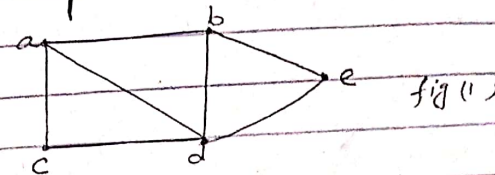
Graph Terminology

Type	Edges	multiple edges Allowed?	Loops Allowed?
Simple graph	undirected	No	No
multigraph	undirected	Yes	No
Pseudograph	undirected	Yes	Yes
Directed graph	directed	No	Yes
Directed multi graph	directed	Yes	Yes

Connectivity

→ A graph is said to be connected if there is a path between every pair of vertex.

Example:-

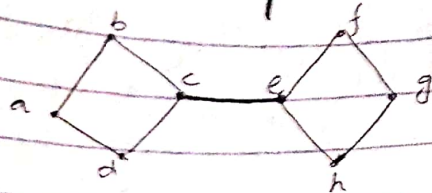


→ fig (2) and fig (3) are said to be disconnected graph. Because there is not a path between every pair of vertex.

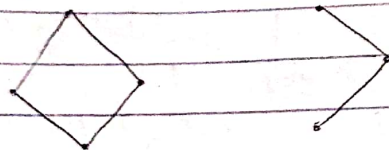
Cut vertex:

→ Let G be a connected Graph. A vertex $v \in G$ is called cut vertex of G

If delete v from ' G '
result in a disconnected graph.
for example:-



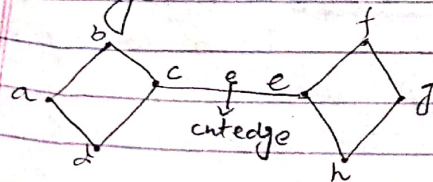
If we remove ' e ' or ' c '
graph will be disconnected



Hence c and e are cut
vertices.

Cut edge:

→ If removing an edge
in a graph result in
two or more graph, then
that edge is called cut
edge.

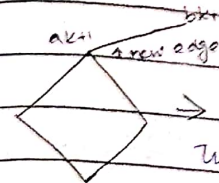


$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2$$

$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2$$

$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2 \quad \text{True}$$

Case II:



→ one of the
two vertices of new
edge is already

$\delta_{k+1} = \delta_k$ edge is already
 $v_{k+1} = v_{k+1}$ in G_k .

$$e_{k+1} = e_{k+1}$$

$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2$$

$$= e_{k+1} - [v_{k+1}] + 2$$

$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2 \quad \text{True}$$

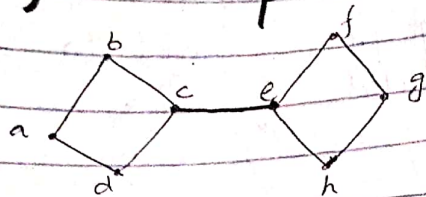
By induction method
Euler's formula is true.

Graph Coloring

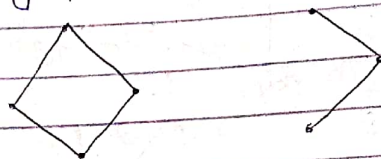
→ Coloring of Graph constitutes
coloring vertices/edges of the
Graph.

→ Coloring all the vertices
of a Graph is the
property that no two vertices

if delete v from " G "
result in a disconnected graph.
for example:-



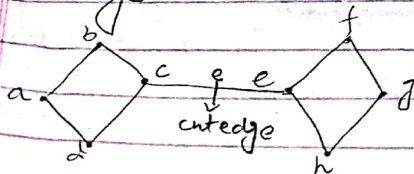
if we remove ' e ' or ' c '
graph will be disconnected



Hence c and e are cut
vertices.

Cut edge:

→ If removing an edge
in a graph result in
two or more graph, then
that edge is called cut
edge.



v_{k+1} are in G_k

Date

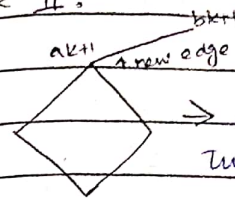
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$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2$$

$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2$$

$$\delta_k = e_k - v_k + 2 \quad \text{True}$$

Case II:



→ one of the
two vertices of new

$\delta_{k+1} = \delta_k$ edge is already

$v_{k+1} = v_{k+1}$ in G_k .

$$e_{k+1} = e_{k+1}$$

$$\delta_{k+1} = e_{k+1} - v_{k+1} + 2$$

$$= e_{k+1} - [v_{k+1}] + 2$$

$$\delta_k = e_k - v_k + 2 \quad \text{True}$$

By induction method

Euler's formula is true.

Graph Coloring

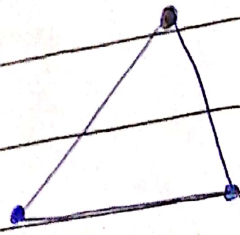
→ Coloring of Graph constitutes
coloring vertices/edges of the
Graph.

→ Coloring all the vertices
of a Graph is the
property that no two vertices

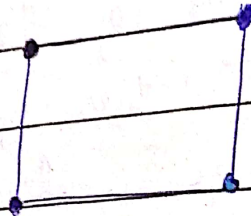
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have same color.

Always try to color
minimum colors.



3-color graph



4-color graph

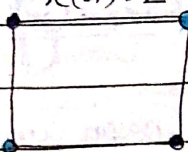
Chromatic numbers:

→ It is defined as the least no of colors needed for coloring the graph.

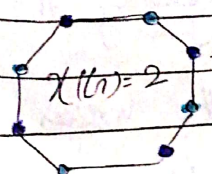
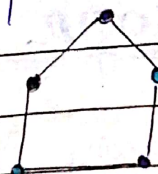
→ denoted by $\chi(G)$ or

k -chromatic graph.

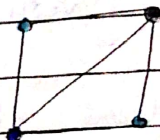
$\chi(G)=2$



$\chi(G)=$



$\chi(G)=2$

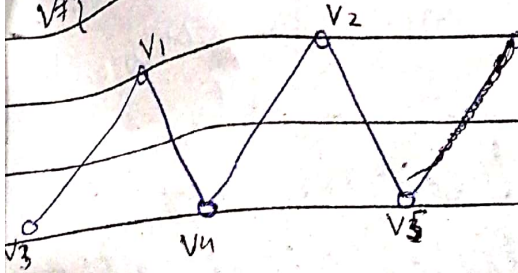


$\chi(G)=3$

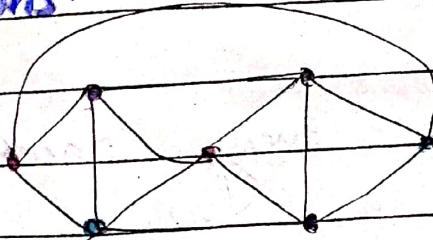
bipartite
 Prove every graph is 2-colorable.

$$\chi(G) = 2$$

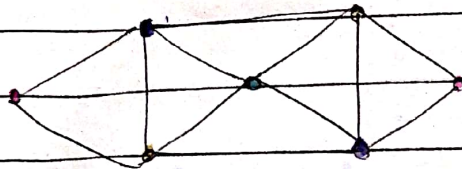
$V = \{V_1, V_2, V_3, V_4, V_5\}$



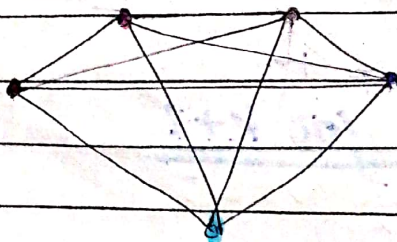
What is the chromatic
 no of the following
 graphs.



$$\chi(G) = 4$$



$$\chi(G) = 3$$



$$\chi(G) = 5$$

5-Color Theorem Proof:

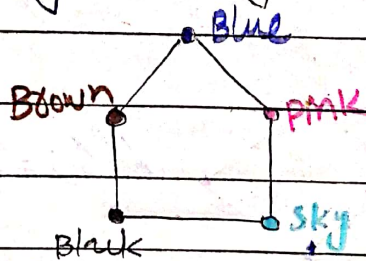
→ Every planar graph with 'n' vertices can be colored using at most 5 colors.

Prove theorem by mathematical induction
Method

i) Basic Step:- $P(n \leq 5) \rightarrow$

Graph can be colored using colors.

Lemma:- Every planar graph contains a vertex with degree $\deg(v) \leq 5$.



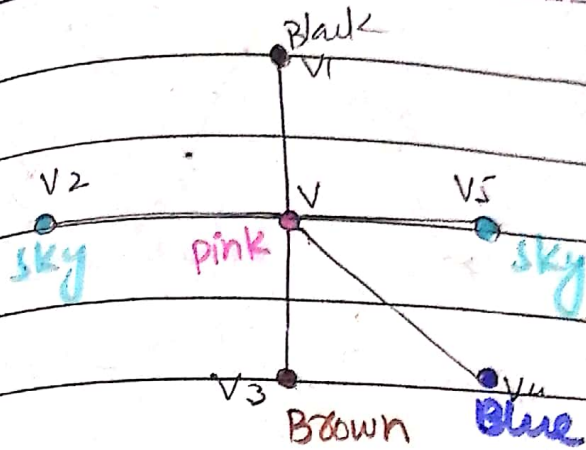
ii) Induction Step:-

Let $k=4$

$\deg(v) \leq 4$

iii) Verifying for $k+1$

$\deg(v) \leq 5$

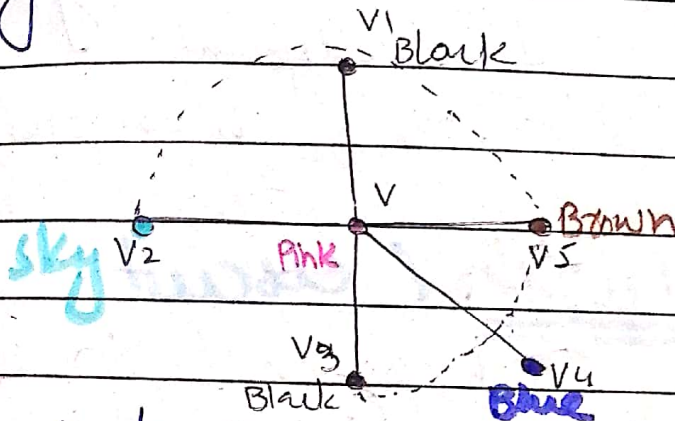


Case 1:-

No disjoint edge

Case 2:-

Assume that there is disjoint edge b/w V_2 and V_5



Let an edge b/w V_4 & V_5

Walk:-

→ A walk is a sequence of vertices and edges of a graph. If we traverse a graph then we get a walk.

Note = Vertices and edges can be repeated.

Graph Colouring

- A planar graph needed four colours.

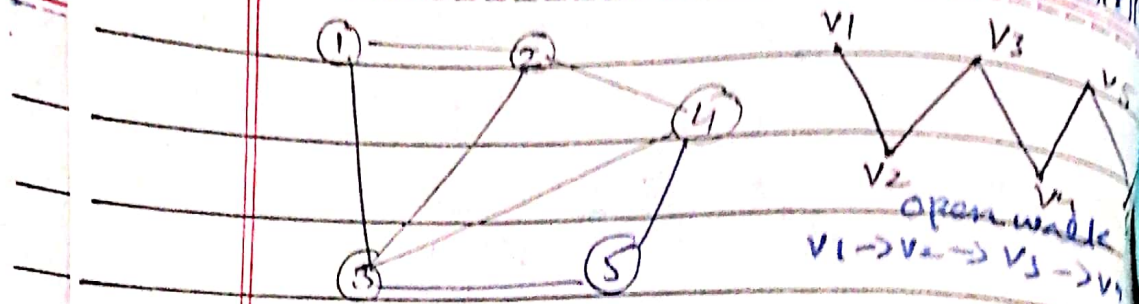
- A K_n graph need n colours.

→ A $K_{m,n}$ graph need 2 colours

→ A C_n need 3 colour when n is odd. and for even $\chi(C_n) = 2$

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Here, $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow 3$
is a walk.

Open walk-path

→ A walk is said to be an open walk if the starting and ending vertices are different.

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 3$ is an open walk.

Closed walk:- / circuit

→ A walk is said to be a closed walk if the starting and ending vertices are identical.

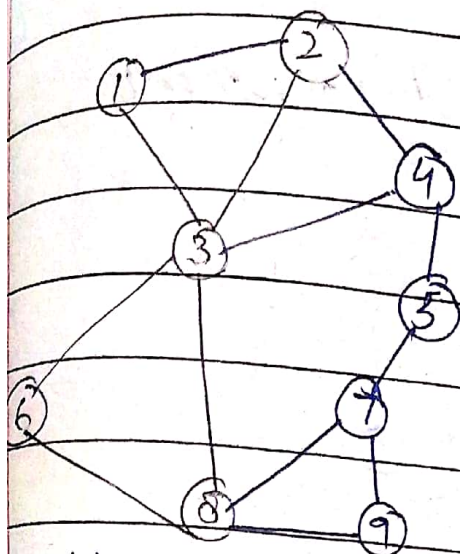
$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 1$ is a closed walk.

Trail:-

Trail is an open walk in which no edge is repeated.

vertex can be repeated.

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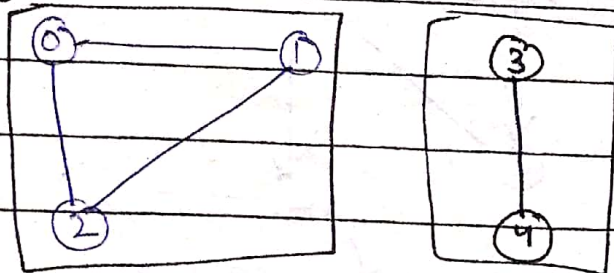
Here, $1 \rightarrow 3 \rightarrow 8 \rightarrow 6 \rightarrow 3 \rightarrow 2$ is trail.

Also $1 \rightarrow 3 \rightarrow 8 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$ will be a closed trail.

→ Connected components of graph

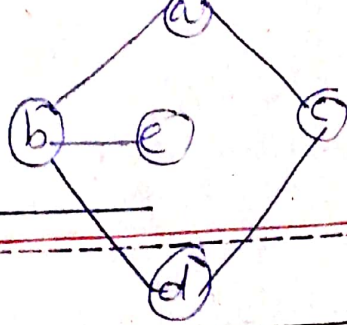
Component of a graph in which every vertex has a path to every other vertex in that component.

connected



2-connected components

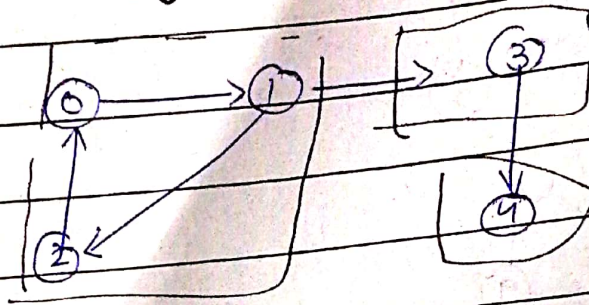
Date _____



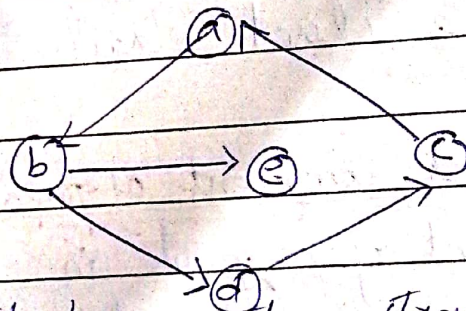
1-connected component

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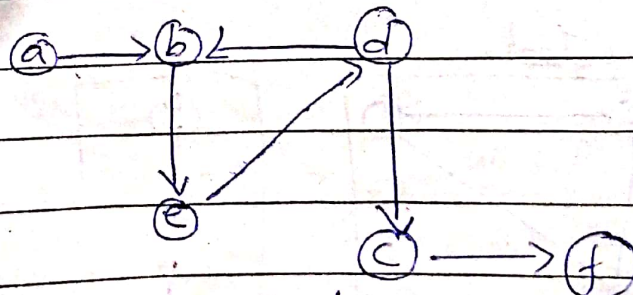
→ Strongly connected components



3- Strongly connected components



It has two strongly connected components.



4- connected components

Strongly connected graph :-

A graph is strongly connected if there is path from a to b and b to a whenever a and b are vertices in the graph.

Weakly connected graph :-

A directed graph is weakly connected if there is path between every two vertices in the underlying undirected graph.

Remember:

1) Walk: vertices may repeat (closed or open)
Edges may repeat

2) Trail: vertices may repeat (open)
Edges cannot repeat.

3) Circuit: vertices may repeat (closed)
edges cannot repeat.

4) Path: vertices cannot repeat.
Edges cannot repeat (open)

5) Cycle: vertices cannot repeat.
Edges cannot repeat (closed)

Write down special cases of complete bipartite graph.

- ★ $K_{1,1} \rightarrow$ Path graph P_2
- ★ $K_{1,2} \rightarrow$ Path graph P_3
- ★ $K_{1,3} \rightarrow$ Claw graph
- ★ $K_{1,n} \rightarrow$ Star graph S_n
- ★ $K_{2,2} \rightarrow$ Square graph C_4
- ★ $K_{3,3} \rightarrow$ Utility graph
- ★

Q# State Dirac's theorem.

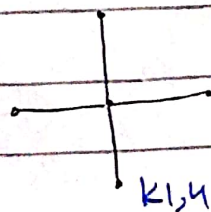
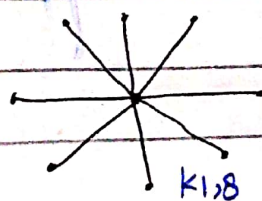
$\Rightarrow$ if G is a simple graph with n vertices, where $n \geq 3$

if $\deg(v) \geq \frac{n}{2}$ for each vertex v , then the graph G is Hamiltonian graph.

Define Star graph.

$\rightarrow$ These have a single central vertex that is connected to n outer vertices and are the same

as: $K_{1,n}$



Q#: Define hypergraph.

→ A graph whose hyper edges connect two or more vertices.

Q#: Define Trivial graph.

→ A graph with no edges and one vertex.

Q#: What is diameter of a graph.

→ Maximum distance between

any two vertices in a graph.

Q#: How many vertices and edges of non-planar graph?

→ A non-planar graph

with minimum number of vertices is K_5 .

→ A non-planar graph with minimum number of

edges is $K_{3,3}$.

Q#: What is a graph theory?

⇒ A graph theory is a

study of graphs in discrete mathematics. The graphs here

are represented by vertices (V)

and edges (E). A graph is symbolised

as $G(V, E)$.

complete graph has $(n-1)!$ circuit

graph.

Step to draw complete graph:-

→ How many vertices n in

your graph?

→ Degree of each vertex $n-1$

→ Find the number of

edges if n is given vertices

$$e = \frac{n(n-1)}{2}$$

→ Draw the complete graph

above values.

* Complete graph (K_1)

number of vertex 2

$$\text{degree of a vertex} = n-1 = 2-1 = 1$$

$$\text{number of edges} = \frac{n(n-1)}{2} = \frac{2(2-1)}{2} = 1$$



* Complete graph (K_2)

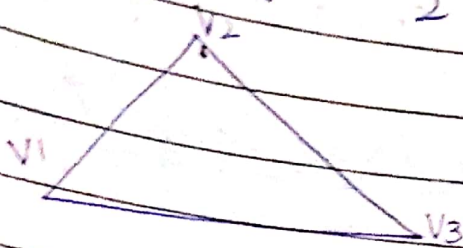
number of vertex 3

$$\text{degree of vertex} = 3-1 = 2$$

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number of edges = $\frac{3(3-1)}{2} = 3$

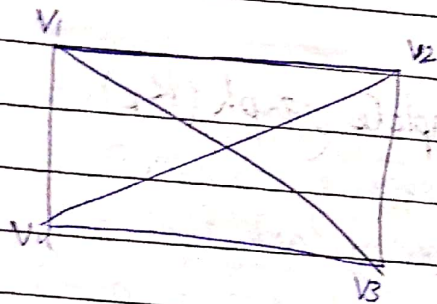


Complete graph (K_3)

number of vertex = 3

degree of its vertex = $3-1=2$

number of edges = $\frac{3(3-1)}{2} = 3$

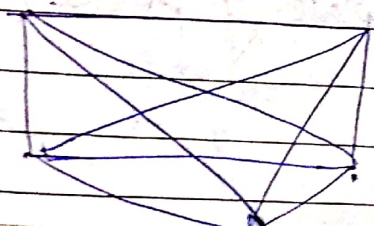


Complete graph 4:-

number of vertex = 4

degree of a vertex = $4-1=3$

edges = $\frac{4(4-1)}{2} = 6$

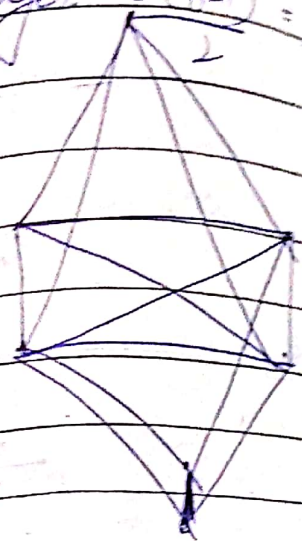


Complete graph (K_5)

number of vertex = 5

degree of vertex = $5-1=4$

edges = $\frac{5(5-1)}{2} = 10$

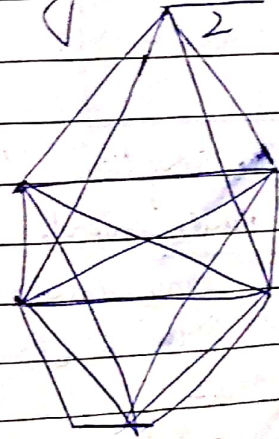


Complete graph (K_6)

number of vertex = 6

degree of vertex = $6-1=5$

edges = $\frac{6(6-1)}{2} = 15$



Rules of Inference

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	Rule of inference	Name	Rule of inference
i)	$\frac{P}{\therefore P}$	Addition	$\frac{P \vee Q}{\neg P}$
	$\frac{VQ}{\therefore Q}$		
ii)	$\frac{P}{Q}$	Conjunction	$\frac{P \rightarrow Q}{Q \rightarrow R}$
	$\therefore P$		$\therefore P \rightarrow R$
	$\wedge Q$		
iii)	$\frac{P \wedge Q}{\therefore P}$	Simplification	$\frac{(P \rightarrow Q) \wedge (R \rightarrow S)}{P \vee R}$
			$\therefore Q \vee S$
iv)	$\frac{P \rightarrow Q}{P}$	modus Phonens	$\frac{(P \rightarrow Q) \wedge (R \rightarrow S) \rightarrow Q}{V \rightarrow S}$
	$\therefore Q$		$\therefore \neg P \vee \neg R$
v)	$\frac{P \rightarrow Q}{\neg Q}$	modus Tollens	
	$\therefore \neg P$		

Q#

How many relations does on a set with n elements have?

Number of symmetric relations on a set with n elements

$$= 2^{n(n+1)/2}$$

QH

How many relations on set $\{1, 2\}$

2

Union of Two graphs

The union of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with the vertex set $V_1 \cup V_2$ and $E_1 \cup E_2$.

The union of two graphs denoted by $G_1 \cup G_2$

Phonens

$$R \rightarrow S) \rightarrow Q$$

$$V \rightarrow S$$

$$\therefore \neg PV \rightarrow R$$